

Alejandro Fontan Villacampa

Research Scientist | 3D Computer Vision & Visual SLAM



[Personal Website](#)

About me

I'm a Visual SLAM Engineer with expertise in Computer Vision, Robotics, AR/VR, and Deep Learning. Passionate about mentorship and collaboration. My career is dedicated to exploring and shaping the future of Visual SLAM and Localization.

Personal

Alejandro Fontan Villacampa
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Areas of specialization

Visual SLAM • Localization
• Autonomous Navigation
• Visual Odometry • VPR

Interests

Computer Vision • Robotics
• Deep Learning • AR/VR
• Embodied AI

Links

[GitHub](#)
 [LinkedIn](#)
 [Scholar](#)
 [@AFontanVillcmp](#)

References

Michael Milford
Javier Civera
Tobias Fischer
Rudolph Triebel
Markus Eder

Other Activities

IELTS Gen. Training (Score 8/9)
First Aid Course & CPR
Professional Music Studies

SHORT RESUMÉ

2022–2027 Research Fellow

QUEENSLAND UNIVERSITY OF TECHNOLOGY · Brisbane, Australia
At the QCR, I lead research on Visual SLAM. Recent highlights include *AllFeature-VSLAM*, a complete framework that tightly integrates diverse features, and a novel *ground-truth-free tuning* method for SLAM/SfM. I also mentor PhD students and our research team in Localization tasks.

2018–2022 PhD Information-Driven Navigation - Associate Researcher

GERMAN AEROSPACE CENTER (DLR) · Munich, Germany
Pioneered *Information-driven direct rgb-d odometry* and *Semi-direct SID-SLAM* to improve visual odometry for mobile and aerial robots in planetary exploration. Deployed and evaluated on a custom DLR multicopter, contributing to its development as a research platform.

2017–2018 Internship as Software Engineer

WIKITUDE GMBH · Salzburg, Austria
Developed a computer vision AR library for mobile OS like Android and iOS, focusing on implementing and evaluating egocentric visual-inertial SLAM algorithms on mobile devices, *OKVIS* and *VINS-Mono*. Created calibration solutions for phone cameras and IMUs' extrinsics. [Wikitude Webpage](#)

LATEST PROJECTS

VSLAM-LAB: A Comprehensive Framework for Visual SLAM Baselines and Datasets.
AllFeature-VSLAM: A Tightly Coupled Multi-Feature Approach for Monocular SLAM
Bommie Toolkit: Automated 3D reconstruction tool for coral reefs
AnyFeature-VSLAM: Automating the Usage of Any Chosen Feature into Visual SLAM

Workshop Organizer:

Mapping-the-Reef@RSS26: 1st Organizer
The-Good-Reviewer@ICRA26: 1st Organizer
Unifying-SLAM@RSS25: 1st Organizer

EDUCATION

2026 **Lecturer for ENN584 SLAM-COURSE** ·
2025 **CV for Spatial Intelligence**
ICVSS · Summer School
2018/22 **PhD System Engineering and Programming**
UNIV. ZARAGOZA · DLR
Advisor: Dr. Javier Civera
Co-Advisor: Dr. Rudolph Triebel
2011/17 **Degree & Master of Industrial Engineering**
EINA · Univ. Zaragoza

MAIN PUBLICATIONS

arXiv 25 *Pixi: Unified Software Development and Distribution for Robotics & AI.*
IROS 25 *VSLAM-LAB: A Comprehensive Framework for Visual SLAM Methods and Datasets.*
arXiv 25 *Look Ma, No Ground Truth! Ground-Truth-Free Tuning of SfM and VSLAM.*
IROS 25 *Image-Based Relocalization and Alignment for Long-Term Monitoring of Dynamic Underwater Environments.*
RSS 24 *AnyFeature-VSLAM: Automating the Usage of Any Feature into Visual SLAM.*
ICRA 24 *Adaptive Outlier Thresholding for Bundle Adjust. in VSLAM.*
BMVC 23 *Motion-Bias-Free Feature-Based SLAM.*
RA-L 23 *SID-SLAM Semi-Direct Information Driven RGB-D SLAM.*
RA-L 22 *A Model for Multi-View Residual Covariances based on Perspective Deformation.*
ICRA 21 *DOT: Dynamic Object Track..*
JFR 21 *The MADMAX data set for visualinertial rover navigation on Mars.*
CVPR 20 *Information-Driven Direct RGB-D Odometry.*

PROGRAMMING

C++ / Python	PyTorch / CUDA	OpenCV / Rerun	ONNX
Anaconda / Pixi	ROS / Isaac Sim	Ceres / g2o / GTSAM	Matlab / $\LaTeX$ / Git

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QUT Brisbane, Australia